

Reconnaissance and Network Enumeration Report: GROUP 8

1. Objective

The objective of this practical was to perform reconnaissance and network enumeration on a vulnerable machine (Metasploitable 2) using Kali Linux tools such as Nmap.

The goal was to identify live hosts, open ports, running services, and potential security weaknesses in a controlled lab environment.

2. Lab Setup

The lab environment consisted of two virtual machines configured in VirtualBox.

Machine 1:

Name: Kali Linux

Role: Attacker machine

IP Address: 10.106.237.120

Machine 2:

Name: Metasploitable 2

Role: Target vulnerable machine

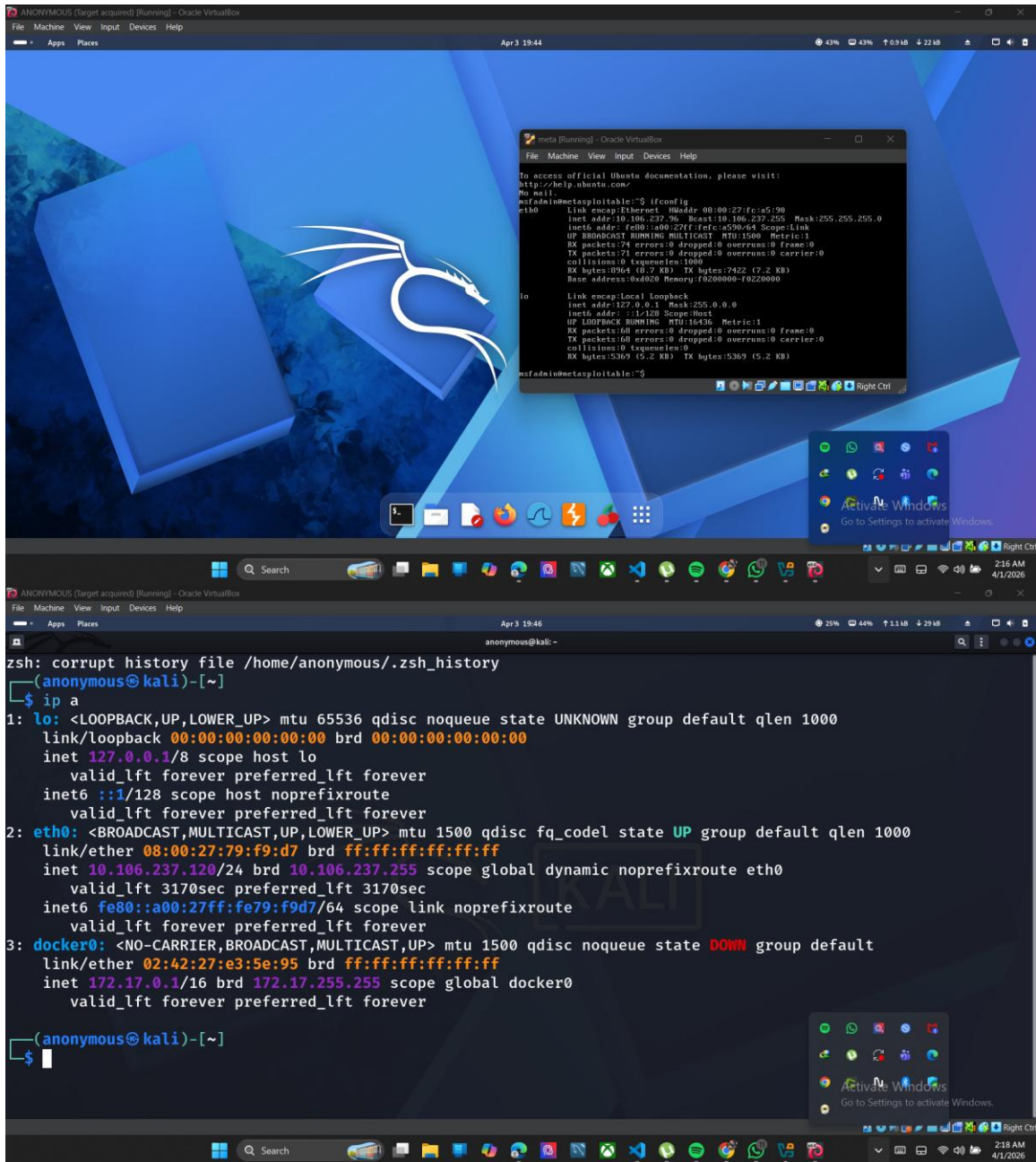
IP Address: 10.106.237.96

Network Range:

10.106.237.0/24

Both machines were configured on the same network to allow communication between attacker and target systems.

SCREENSHOT 1 – IP configuration



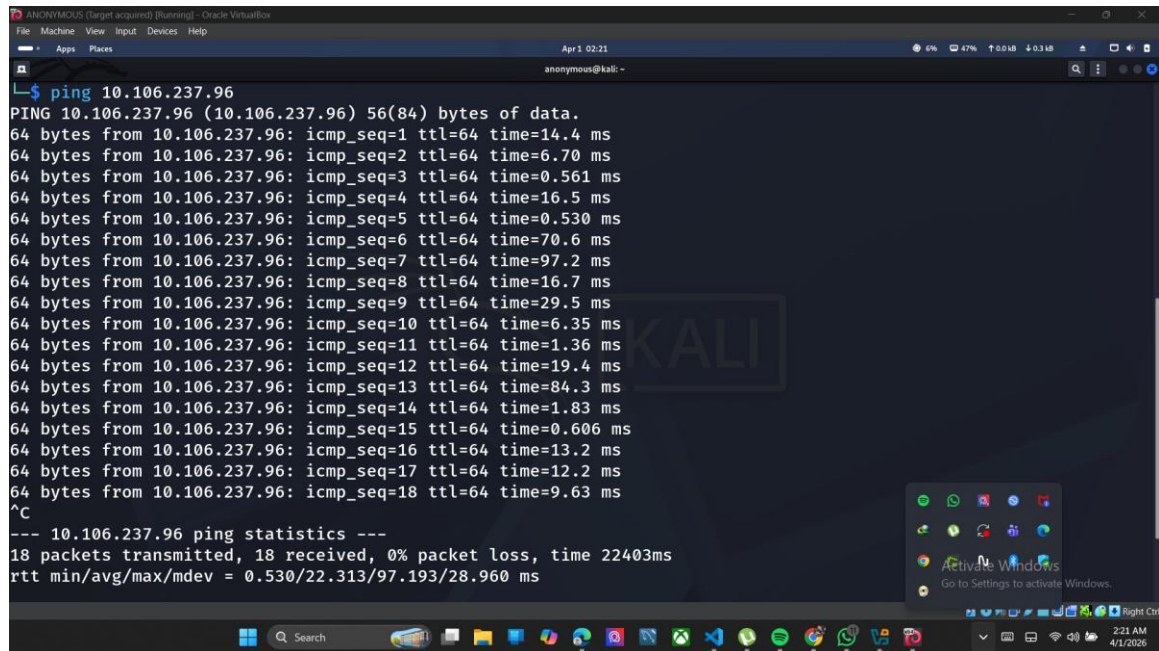
3. Connectivity Test

Command used:

ping 10.106.237.96

The successful ping response confirmed that the Kali machine could communicate with the Metasploitable machine across the network.

SCREENSHOT 2 – Ping result



The screenshot shows a Kali Linux terminal window with the following output:

```
anonymous@kali: ~$ ping 10.106.237.96
PING 10.106.237.96 (10.106.237.96) 56(84) bytes of data:
64 bytes from 10.106.237.96: icmp_seq=1 ttl=64 time=14.4 ms
64 bytes from 10.106.237.96: icmp_seq=2 ttl=64 time=6.70 ms
64 bytes from 10.106.237.96: icmp_seq=3 ttl=64 time=0.561 ms
64 bytes from 10.106.237.96: icmp_seq=4 ttl=64 time=16.5 ms
64 bytes from 10.106.237.96: icmp_seq=5 ttl=64 time=0.530 ms
64 bytes from 10.106.237.96: icmp_seq=6 ttl=64 time=70.6 ms
64 bytes from 10.106.237.96: icmp_seq=7 ttl=64 time=97.2 ms
64 bytes from 10.106.237.96: icmp_seq=8 ttl=64 time=16.7 ms
64 bytes from 10.106.237.96: icmp_seq=9 ttl=64 time=29.5 ms
64 bytes from 10.106.237.96: icmp_seq=10 ttl=64 time=6.35 ms
64 bytes from 10.106.237.96: icmp_seq=11 ttl=64 time=1.36 ms
64 bytes from 10.106.237.96: icmp_seq=12 ttl=64 time=19.4 ms
64 bytes from 10.106.237.96: icmp_seq=13 ttl=64 time=84.3 ms
64 bytes from 10.106.237.96: icmp_seq=14 ttl=64 time=1.83 ms
64 bytes from 10.106.237.96: icmp_seq=15 ttl=64 time=0.606 ms
64 bytes from 10.106.237.96: icmp_seq=16 ttl=64 time=13.2 ms
64 bytes from 10.106.237.96: icmp_seq=17 ttl=64 time=12.2 ms
64 bytes from 10.106.237.96: icmp_seq=18 ttl=64 time=9.63 ms
^C
--- 10.106.237.96 ping statistics ---
18 packets transmitted, 18 received, 0% packet loss, time 22403ms
rtt min/avg/max/mdev = 0.530/22.313/97.193/28.960 ms
```

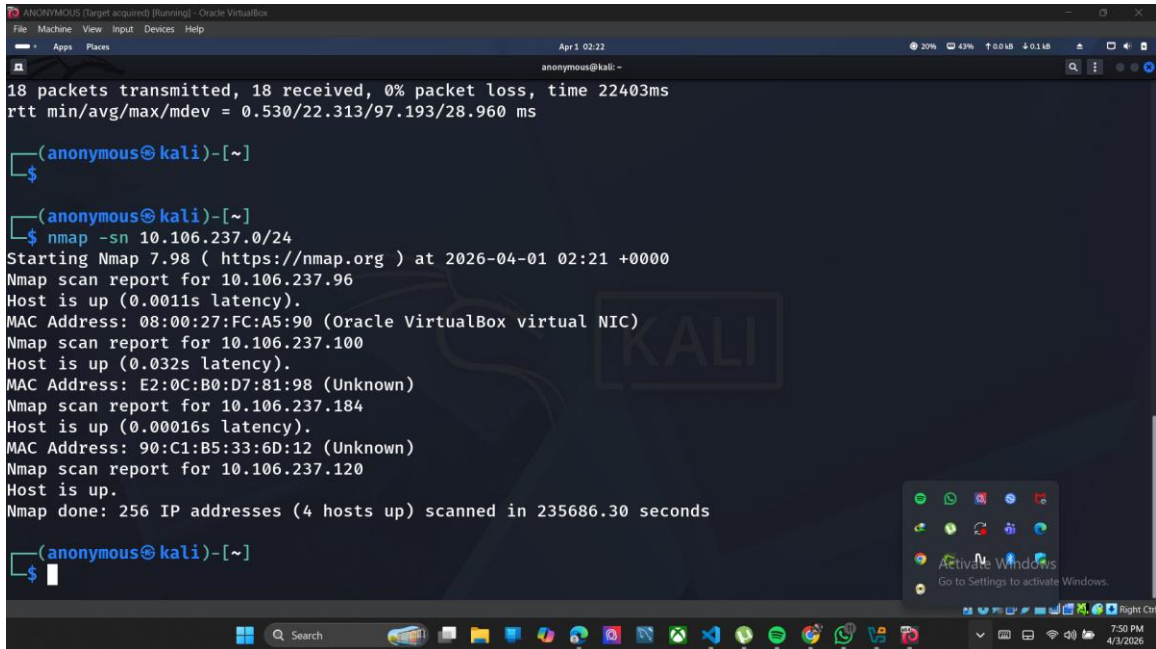
4. Host Discovery

Command used:

```
nmap -sn 10.106.237.0/24
```

This scan was used to identify active hosts within the network. The Metasploitable machine was detected as an active host.

SCREENSHOT 3 – Host discovery



The screenshot shows a Kali Linux terminal window with the following text:

```
18 packets transmitted, 18 received, 0% packet loss, time 22403ms
rtt min/avg/max/mdev = 0.530/22.313/97.193/28.960 ms

(anonymous@kali)-[~]
$

(anonymous@kali)-[~]
$ nmap -sn 10.106.237.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 02:21 +0000
Nmap scan report for 10.106.237.96
Host is up (0.0011s latency).
MAC Address: 08:00:27:FC:A5:90 (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.106.237.100
Host is up (0.032s latency).
MAC Address: E2:0C:B0:D7:81:98 (Unknown)
Nmap scan report for 10.106.237.184
Host is up (0.00016s latency).
MAC Address: 90:C1:B5:33:6D:12 (Unknown)
Nmap scan report for 10.106.237.120
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 235686.30 seconds

(anonymous@kali)-[~]
$
```

The terminal window is titled "ANONYMOUS [target acquired] [running] - Oracle VirtualBox". The desktop background is dark with a "KALI" logo. The taskbar at the bottom shows various application icons and the system clock indicates 7:50 PM on 4/3/2026.

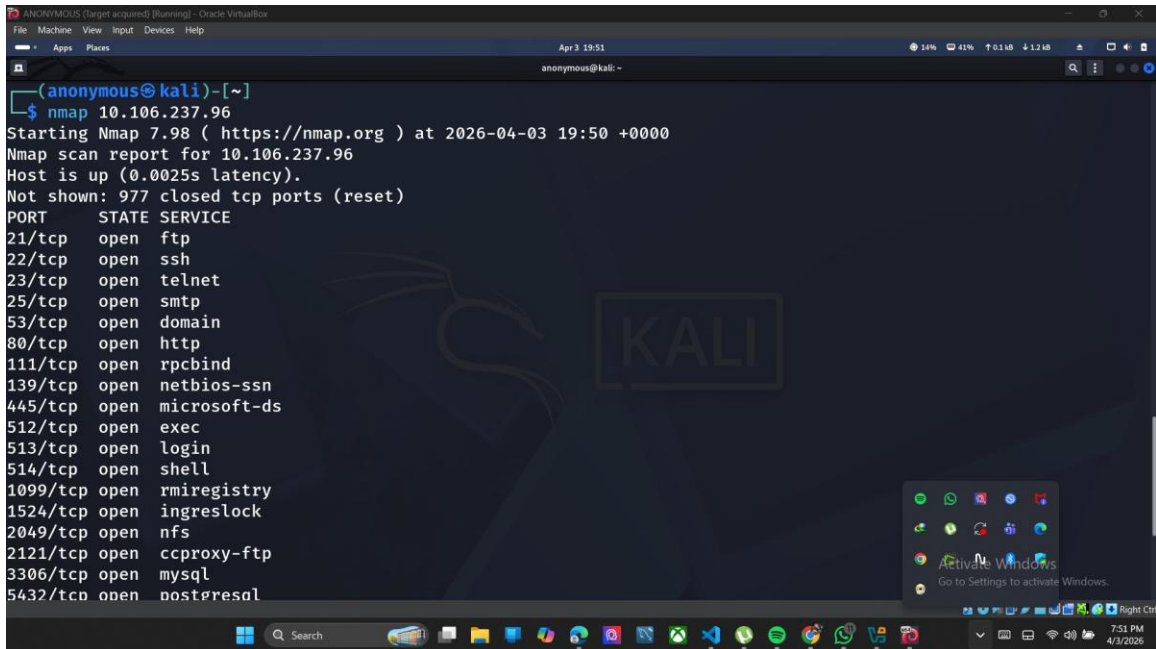
5. Port Scanning

Command used:

`nmap 10.106.237.96`

The scan revealed multiple open ports indicating exposed services on the target system.

SCREENSHOT 4 – Open ports



```
(anonymous@kali)-[~]
$ nmap 10.106.237.96
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 19:50 +0000
Nmap scan report for 10.106.237.96
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
```

6. Service Enumeration

Command used:

`nmap -sV 10.106.237.96`

Service enumeration identified the versions of services running on open ports.

Examples of identified services:

FTP service (vsftpd)

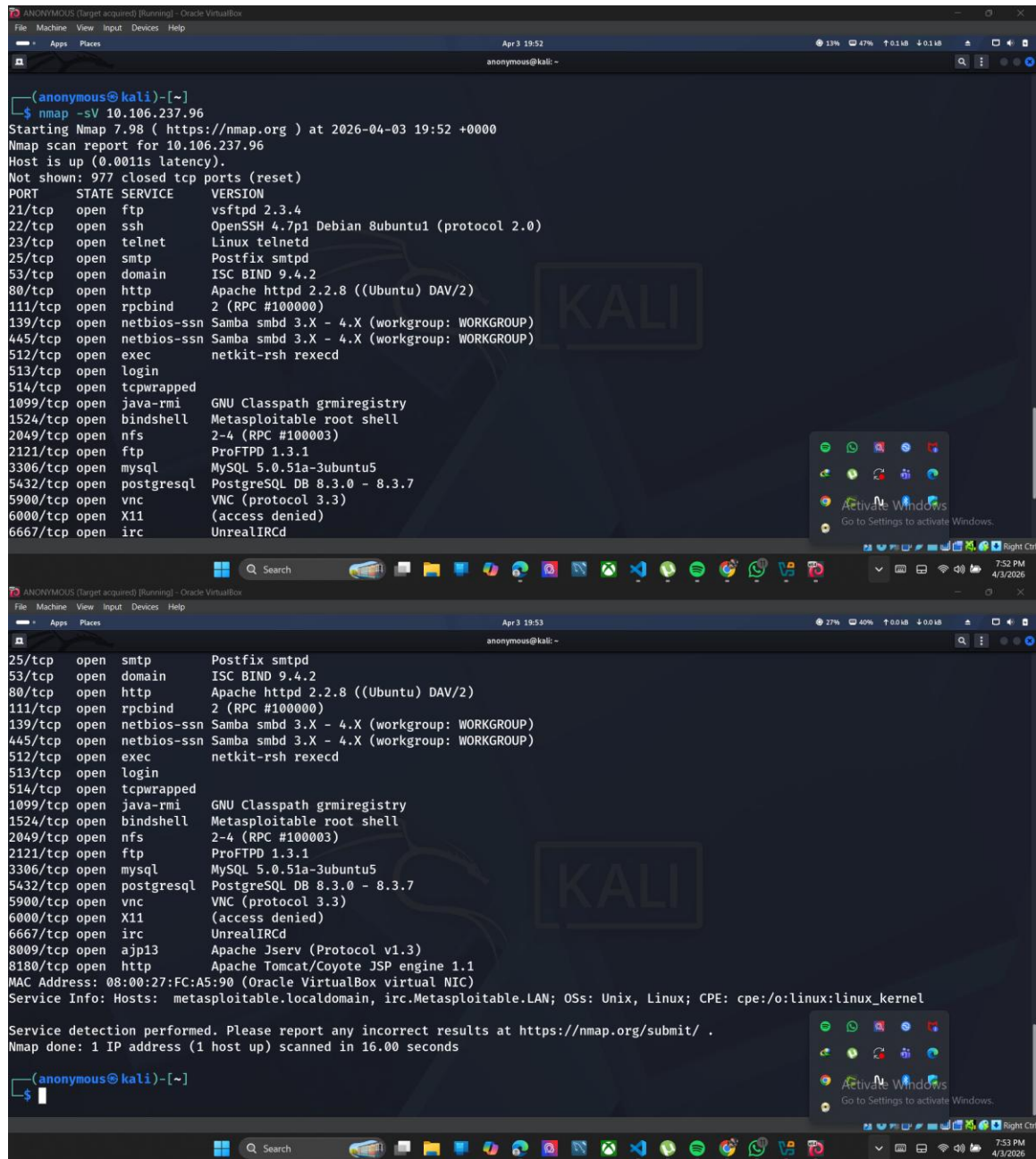
SSH service

Apache web server

MySQL database

Telnet service

SCREENSHOT 5 – Service versions



```
(anonymous@kali)-[~]
$ nmap -sV 10.106.237.96
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 19:52 +0000
Nmap scan report for 10.106.237.96
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd

25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:FC:A5:90 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.00 seconds

(anonymous@kali)-[~]
$
```

7. Aggressive Scan

Command used:

```
sudo nmap -A 10.106.237.96
```

The aggressive scan provided additional information including:

Operating system details

Service configurations

Additional script results

SCREENSHOT 6 – Aggressive scan

```
anonymous@kali:~$ sudo nmap -A 10.106.237.96
[sudo] password for anonymous:
Sorry, try again.
[sudo] password for anonymous:
Sorry, try again.
[sudo] password for anonymous:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 19:56 +0000
Nmap scan report for 10.106.237.96
Host is up (0.035s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_ 1024 60:0f:cf:e1:c0:5f:0a:74:d6:90:24:fa:c4:d5:ec:cd (DSA)
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-04-03T19:55:08+00:00; +2s from scanner time.
|_sslv2:
|_ SSLV2 supported
|_ ciphers:
|_ SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ SSL2_DES_64_CBC_WITH_MD5
|_ SSL2_RC4_128_EXPORT40_WITH_MD5
|_ SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ SSL2_RC4_128_WITH_MD5
|_ SSL2_RC2_128_CBC_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OC05A/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_smtp-comands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_ dns-nsid:
|_ bind-version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-title: Metasploitabl2 - Linux
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_ rpcinfo:
|_ program version port/proto service
|_ 100000 2 111/tcp rpcbind
|_ 100000 2 111/udp rpcbind
|_ 100003 2,3,4 2049/tcp nfs
|_ 100003 2,3,4 2049/udp nfs
|_ 100005 1,2,3 42686/udp mountd
|_ 100005 1,2,3 50986/tcp mountd
|_ 100021 1,3,4 30815/udp nlockmgr
|_ 100021 1,3,4 39305/tcp nlockmgr
|_ 100024 1 42540/udp status
|_ 100024 1 55192/tcp status
139/tcp   open  netbios-ssn Samba smb 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smb 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath gmiregistry
1524/tcp  open  bindshell    Metasploitabl root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.2-1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_ mysql-info:
|_ Protocol: 10
|_ Version: 5.0.51a-3ubuntu5
|_ Thread ID: 9
|_ Capabilities flags: 43564
|_ Some Capabilities: SwitchToSSLAfterHandshake, ConnectWithDatabase, LongColumnFlag, SupportsTransactions, Support41Auth, Speaks41ProtocolNew, SupportsCompression
|_ Status: Autocommit
|_ Salt: j)NDA98N_sE7LAA'cIiU
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
|_ ssl-date: 2026-04-03T19:55:07+00:00; +1s from scanner time.
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OC05A/stateOrProvinceName=There is no such thing outside US/countryName=XY
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
5900/tcp  open  vnc          VNC (protocol 3.3)
|_ vnc-info:
|_ Protocol version: 3.3
|_ Security types:
|_ VNC Authentication (2)
6080/tcp  open  x11          (access denied)
6667/tcp  open  irc          UnrealIRCd
```

```
ANONYMOUS (Target acquired) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Apr 3 19:56
anonymous@kali: -

|_ VNC Authentication (2)
6000/tcp open  x11      (access denied)
6667/tcp open  irc       UnrealIRCd
8009/tcp open  ajp13     Apache Jserv (Protocol v1.3)
|_ ajp-methods: failed to get a valid response for the OPTIONS request
8100/tcp open  http      Apache Tomcat/Coyote JSP engine 1.1
|_ http-server-header: Apache-Coyote/1.1
|_ http-favicon: Apache Tomcat
|_ http-title: Apache Tomcat/5.5
MAC Address: 08:00:27:FC:A5:90 (Oracle VM VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-04-03T15:54:55-04:00
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: mean: 1h00m01s, deviation: 2h00m01s, median: 0s

TRACEROUTE
HOP RTT ADDRESS
1 34.79 ms 10.106.237.96

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.87 seconds

anonymous@kali:~$
```

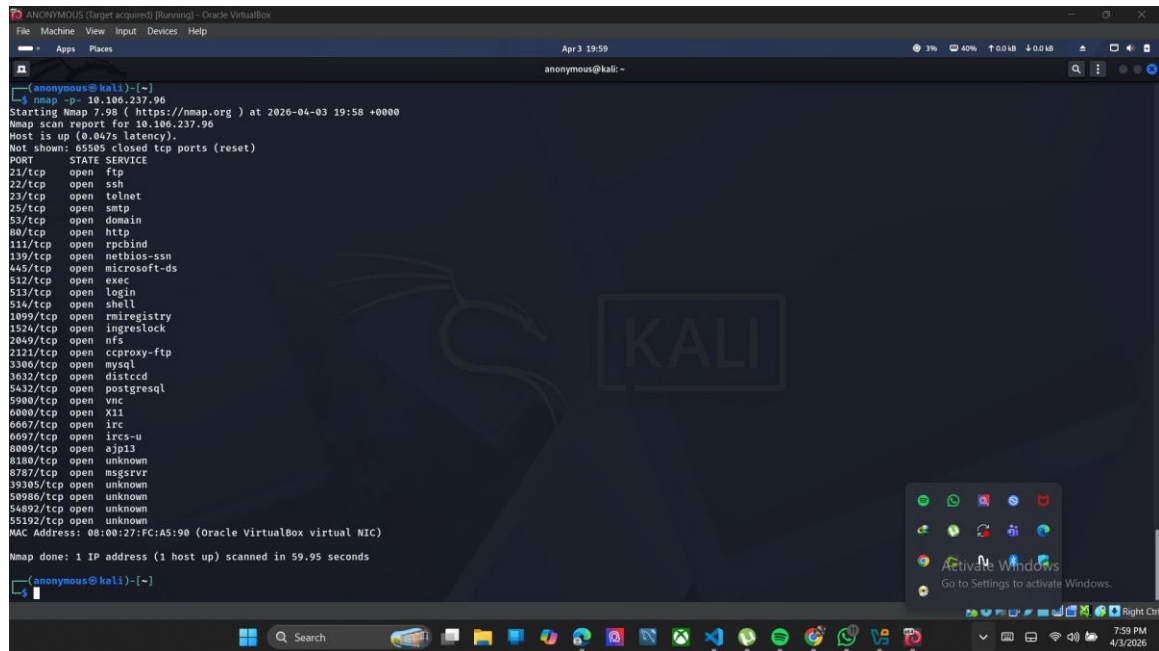
8. Full Port Scan

Command used:

```
nmap -p- 10.106.237.96
```

This scan checked all 65535 ports to identify additional exposed services.

SCREENSHOT 7 – Full port scan



```
anonymous@kali:~$ nmap -p- 10.106.237.96
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 19:58 +0000
Nmap scan report for 10.106.237.96
Host is up (0.047s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6067/tcp  open  irc
6097/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
39809/tcp open  unknown
50986/tcp open  unknown
54892/tcp open  unknown
55192/tcp open  unknown
MAC Address: 08:00:27:FC:A5:90 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 59.95 seconds

anonymous@kali:~$
```

9. Default Script Scan

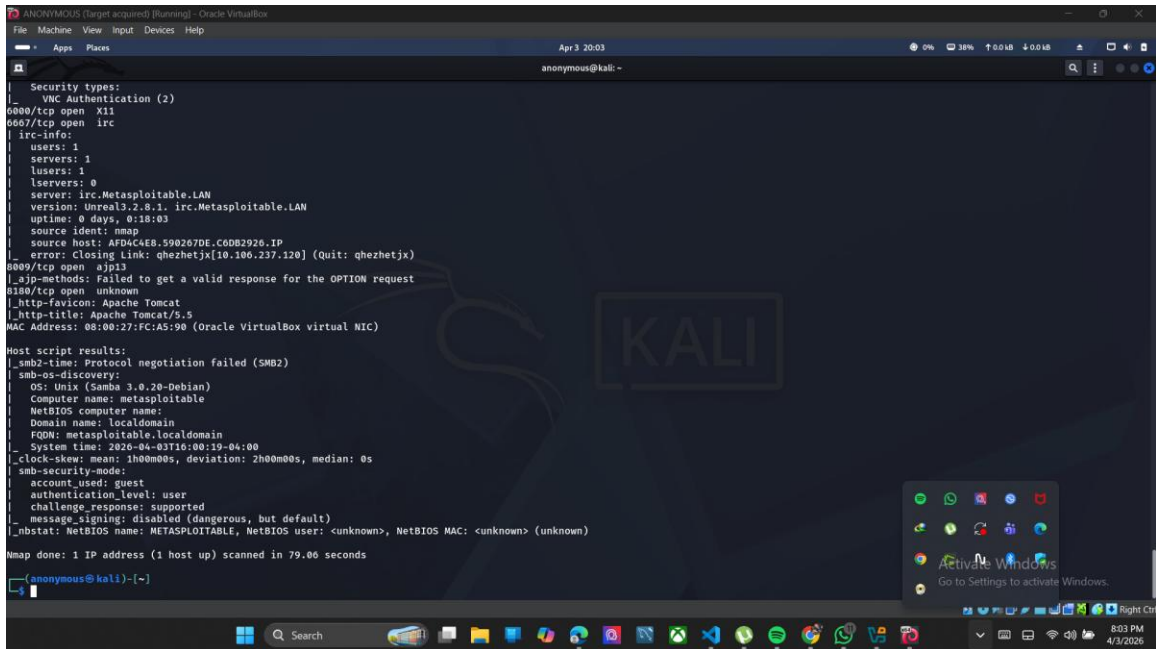
Command used:

`nmap -sC 10.106.237.96`

Default scripts were used to gather more information about services and potential vulnerabilities.

SCREENSHOT 8 – Script scan

```
ANONYMOUS [Target acquired] [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Apr 3 20:02
anonymous@kali: ~
[anonymous@kali]~$ nmap -sC 10.106.237.90
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 20:00 +0000
Nmap scan report for 10.106.237.90
Host is up (0.038s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 10.106.237.120
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:0a:74:d0:90:24:fa:c4:d5:0c:cd (DSA)
|_  2048 56:16:24:0f:21:1d:de:a7:2b:ae:01:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet
25/tcp    open  smtp
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_  SSL2_DES_192_EDE3_CBC_WITH_MD5
|_  SSL2_RC2_128_CBC_WITH_MD5
|_  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_  SSL2_DES_64_CBC_WITH_MD5
|_  SSL2_RC4_128_EXPORT40_WITH_MD5
|_  SSL2_RC4_128_WITH_MD5
|_ssl-cert: Subject: commonName=subuntu894-base.localdomain/organizationName=OC05A/stateOrProvinceName=There is no such thing outside US/countryName=X
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-10T14:07:45
|_ssl-date: 2026-04-03T20:01:19+00:00; 0s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
53/tcp    open  domain
|_dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http
|_http-title: Metasploitale2 - Linux
111/tcp   open  rpcbind
|_rpcinfo:
|_  program version  port/proto  service
|_  100000 2 111/tcp  rpcbind
|_  100000 2 111/udp  rpcbind
|_  100003 2,3,4 2049/tcp  nfs
|_  100003 2,3,4 2049/udp  nfs
|_  100005 1,2,3 42686/udp mountd
|_  100005 1,2,3 50986/tcp mountd
|_  100021 1,3,4 30815/udp nlockmgr
|_  100021 1,3,4 39305/tcp nlockmgr
|_  100024 1 42540/udp status
|_  100024 1 55192/tcp status
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
|_mysql-info:
|_  Protocol: 10
|_  Version: 5.0.51a-3ubuntu5
|_  Thread ID: 20
|_  Capabilities flags: 43564
|_  Some Capabilities: Support41Auth, ConnectWithDatabase, SupportsTransactions, LongColumnFlag, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, Supp
|_  Status: Autocommit
|_  Salt: z{ym:t}Re$*o7bga/f4
5432/tcp  open  postgresql
|_ssl-cert: Subject: commonName=subuntu894-base.localdomain/organizationName=OC05A/stateOrProvinceName=There is no such thing outside US/countryName=X
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-10T14:07:45
|_ssl-date: 2026-04-03T20:01:19+00:00; 0s from scanner time.
5900/tcp  open  vnc
```



10. Attack Surface Summary

The attack surface of the Metasploitable machine includes multiple exposed services which could be targeted by attackers.

FTP (Port 21)

Allows file transfer. May allow anonymous login.

SSH (Port 22)

Used for remote administration. Vulnerable to brute force attacks.

Telnet (Port 23)

Transmits login credentials in plain text making it insecure.

HTTP (Port 80)

Web server which could contain web application vulnerabilities.

SMB (Port 445)

File sharing service commonly targeted by malware.

MySQL (Port 3306)

Database service which may expose sensitive information.

11. Conclusion

The reconnaissance and enumeration process successfully identified active hosts, open ports, and running services on the Metasploitable system.

This demonstrates how attackers gather critical information before launching attacks. Understanding these techniques helps cybersecurity professionals identify and secure exposed services.